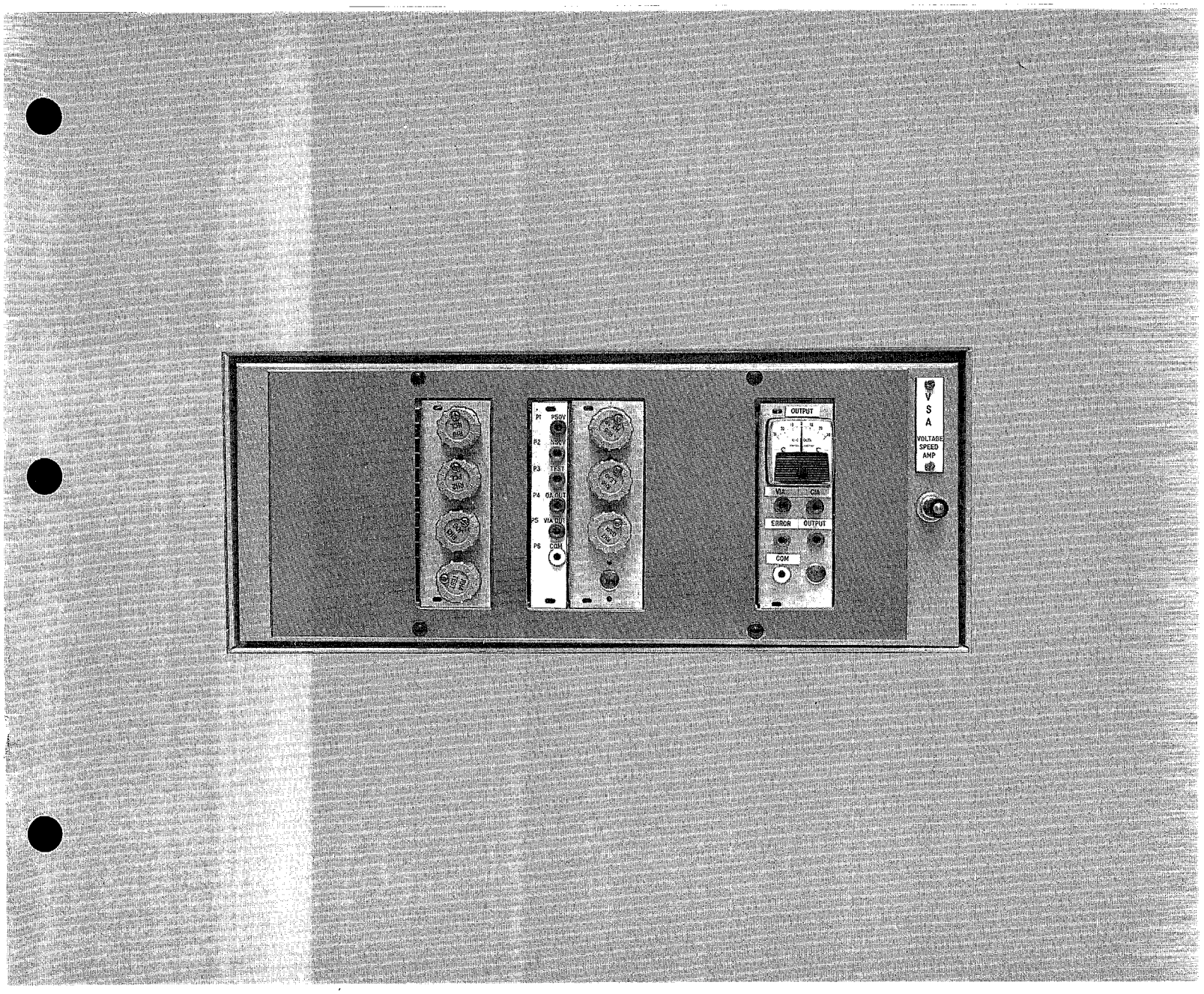
INSTRUCTIONS

VSA
VOLTAGE OR SPEED AMPLIFIER



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GENERAL ELECTRIC

CONTENTS

	Page
INTRODUCTION	3
Test Circuits	3
Amplifier Specifications	3
Speed Regulator Specifications	3
Voltage Regulator Specifications	3
DESCRIPTION	3
OPERATION	5
General	5
Speed Regulator	5
Voltage Regulator	6
Operating Amplifier Card	6
Isolation Amplifier	8
Input Circuit	8
Amplifier Feedback Circuit	9
Current Limit Circuit	9
Meter and Test Features	10
INSTALLATION	10
Installation Adjustments	10
Selection of Voltage or Speed Regulation	10
Maximum Speed Adjustment	10
Tune-up Jumpers	11
Current Limit Adjustment	11
Jog Adjustment	11
Voltage Isolation Amplifier Gain Adjustment	11
Gain Adjustment	11
Lead Adjustment	11
TROUBLESHOOTING	11
General	11
Test Points	12
Procedure	12
APPENDIX A - ISOLATION AMPLIFIER (IC4501A101A)	16
APPENDIX B - OPERATIONAL AMPLIFIER CARD (IC3600A0AC1)	20

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

VSA VOLTAGE OR SPEED AMPLIFIER

INTRODUCTION

The VSA voltage or speed amplifier regulates the armature voltage or shaft speed of a d-c motor where the motor armature power is furnished by a full-regenerating Silcomatic power supply. Typical applications for VSA are mechanical winder drives, windup drives on regenerative winders, and supercalender stack drives. The selection of voltage or speed regulation is made by screw-connected wire jumpers within the VSA voltage or speed amplifier. The VSA also furnishes jog reference, current limit, and various modes of preconditioning depending on the drive requirements.

An input reference voltage to the VSA is compared with the motor voltage or motor tachometer voltage; the difference or error voltage, is amplified by the VSA to produce an output voltage to the Silcomatic power supply. The VSA contains the adjustments required to set the desired operating levels and the components necessary to stabilize the type of regulator selected.

TEST CIRCUITS

The VSA has a test card which is removed from one receptacle and reinserted in the test slot. Test points mounted on the test card allow checks to be performed on the equipment on a go/no-go basis.

AMPLIFIER SPECIFICATIONS

1. Power Input	1. Positive 50 volts, 60 milliamperes 2. Negative 50 volts, 70 milliamperes 3. 250 volts, 182 ma through 1100Ω resistor 4. 250 volts, 55 ma through 4kΩ resistor
2. Input Reference	0 to -29 volts at 0.7 ma for maximum output. Suicided 3. 3K input.
3. Jog Reference	0 to 10 percent of regulated voltage, 1 second time delay
4. Amplifier Limit on Output	±26.5 volts
5. Current Limit Adjustment	0 to 250 percent of full-load motor current

SPEED REGULATOR SPECIFICATIONS

1. Tachometer voltage range (maximum)	82 to 388 volts at 0.7 ma
2. Maximum rate of change of speed amplifier output	11 seconds from zero to maximum output
3. Speed regulator Crossover	0.5 to 2 radians per second with gain adjust
4. Inertia Lead	0.4 to 4.5 seconds with lead adjust
5. Drift (tachometer voltage)	±0.01 to ±0.06 percent per 10 C
6. Offset (tachometer voltage)	0 to ±0.6 percent depending on gain adjust

VOLTAGE REGULATOR SPECIFICATIONS

1. Drift (Silcomatic voltage)	0.02 percent per 10 C
2. Offset (Silcomatic voltage)	0 to 0.2 percent

DESCRIPTION

All components of the VSA voltage or speed amplifier are mounted inside an 8-inch high by 20-inch wide steel enclosure, commonly called a module. This module is designed to mount on control panels where needed in a paper mill drive. Four mounting bolts which provide for convenient removal and replacement secure the module to the panel. All wires entering and leaving the module are connected to screw-type terminals accessible from the front when the cover plate is removed.

Those components visible from the front of the amplifier under operating conditions are shown in Fig. 1. When the front cover plate is removed, front access is obtained to the terminal boards and printed-wiring boards. These components are identified in Fig. 2.

A hinged door with latch provides access to all wiring within the enclosure as shown in Fig. 3. The printed-wiring board pins plug into a matching receptacle mounted at the rear of the door assembly. All interconnecting wires between the receptacle pins of printed boards are installed by means of a gastight pressure connection utilizing a special pressure gun as a tool. This tool simultaneously strips the insulation from the conductor, attaches the lug, and compresses the lug and wire to the pin with a wipe-clean action. The conductor is held, under high pressure, against the surface of the connector pin to provide a gastight connection between conductor and pin.

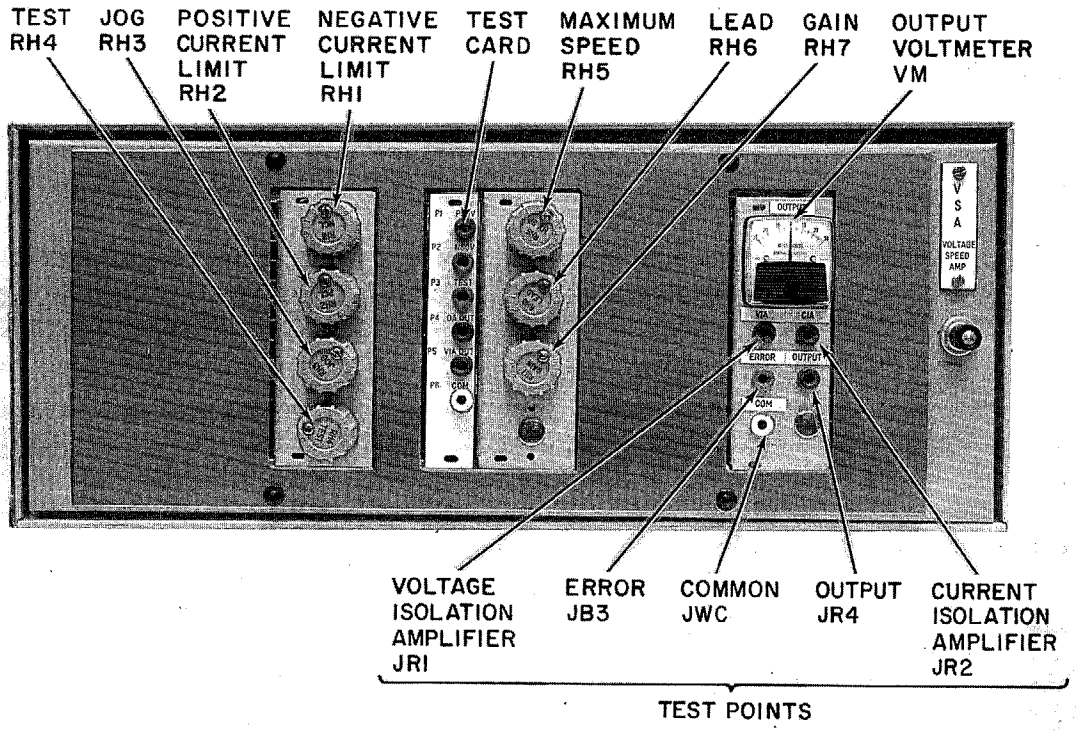


Fig. 1. Front view of VSA voltage or speed amplifier

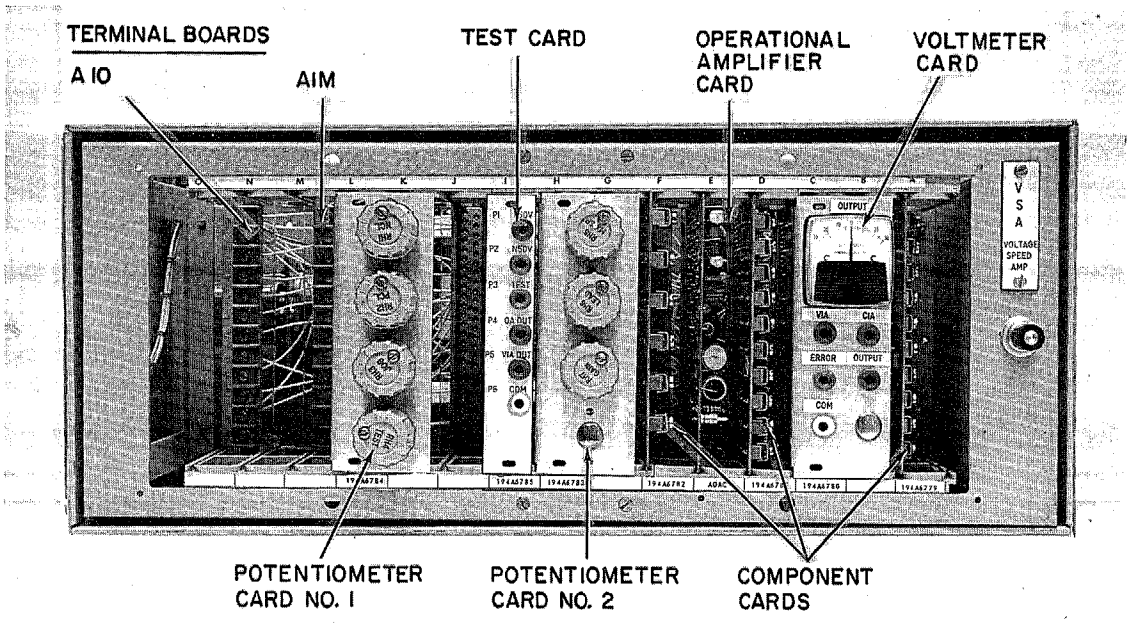


Fig. 2. Front view of VSA voltage or speed amplifier with cover plate removed

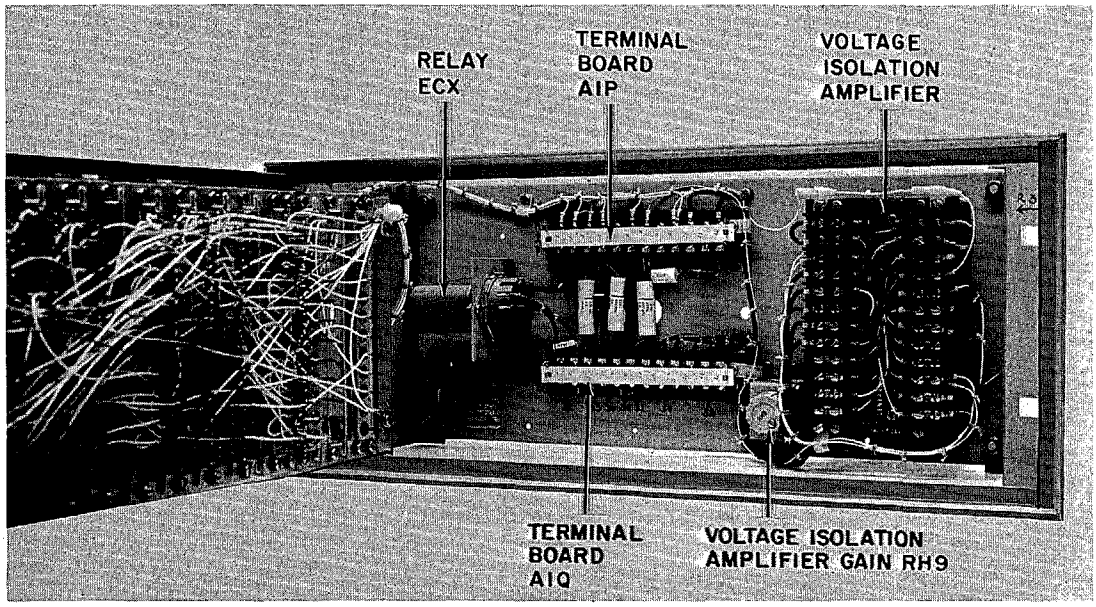


Fig. 3. VSA voltage or speed amplifier with door open, showing internally mounted devices

The printed-wiring boards are made of glass epoxy to resist moisture absorption and warping in mill environments. After the board components are assembled and soldered, the entire board is conformal coated to protect it against dirt, moisture, fumes and other environmental contamination. When the boards are removed, a special card puller supplied with the equipment releases the printed board pins from the receptacle during removal. During replacement, the back of this puller helps to seat the board pins into the mating receptacle. Both the board and receptacle contact pins are nickel-tin plated for good electrical contact and corrosion resistance.

TABLE 1

DEVICE LOCATION AND IDENTIFICATION

Slot Designation	Device Name	Identification No.
A1O	Terminal Board A1O	None
A1N	Not used	----
A1M	Terminal Board A1M	None
A1L	Potentiometer Card No. 1	194A6784G1
A1K	Not used	----
A1J	Not used (Test only)	----
A1I	Test Card	194A6785G1
A1H	Potentiometer Card No. 2	194A6783G1
A1G	Not used	----
A1F	Component Card	194A6782G1
A1E	Operational Amplifier	IC3600A0AC1
A1D	Component Card	194A6781G1
A1C	Voltmeter Card	194A6780G1
A1B	Not used	----
A1A	Component Card	194A6779G1

OPERATION

GENERAL

The VSA voltage or speed amplifier contains two basic amplifiers: an operational amplifier mounted on a plug-in card and an isolation amplifier bolted to the rear of the enclosure. These amplifiers are shown in Fig. 2 and 3. There are seven additional plug-in cards included with the amplifiers. These cards contain the passive components required for the circuits which control the outputs of the amplifiers.

SPEED REGULATOR

Figure 4 shows a simplified block diagram of the VSA module used as a speed regulator. In the speed regulation mode, the MX relay is energized and the input circuit compares tachometer voltage with the reference input and produces an error output which is amplified by the operational amplifier to drive the Silcomatic power supply. The gain of the operational amplifier is set by the amplifier feedback circuit for the desired regulator stability. The current limit circuit, if used, receives its input from a current isolation amplifier which senses motor armature current from a shunt. Whenever the motor armature current exceeds a preset value, the current limit circuit output overrides the speed error from the input circuit and maintains the motor armature current at the preset value.

For jog operation, the reference input is maintained at zero and the jog input sets the desired level of tachometer voltage.

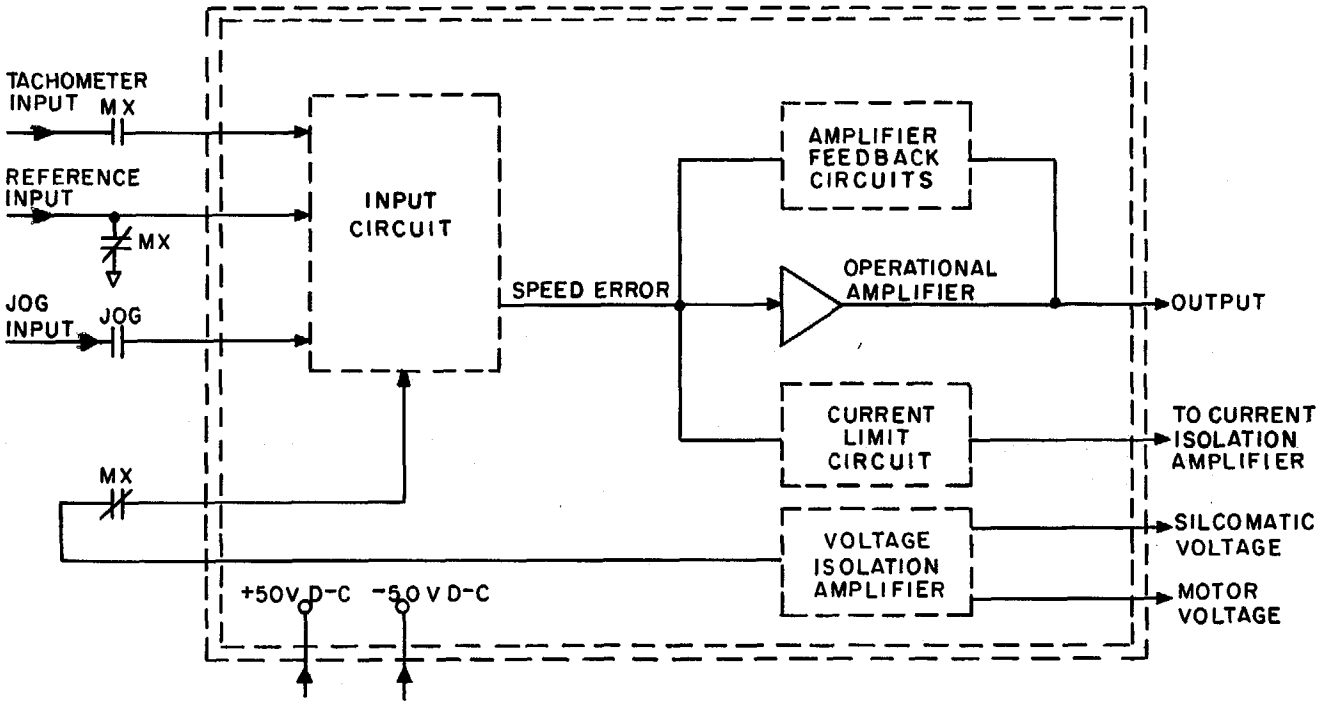


Fig. 4. Simplified diagram of speed regulator

When the VSA voltage or Speed Amplifier is not regulating tachometer voltage, the line contactor connecting the motor armature to the Silcomatic power supply is open and the MX relay is de-energized. Under this condition, the operational amplifier is preconditioned by two methods. One method is shown in Fig. 4 where the voltage isolation amplifier senses the difference between motor voltage and Silcomatic voltage with the operational amplifier regulating this difference to zero. This type of operation is called switch volts regulation and maintains the voltage across the line contactor at zero. The other method is to eliminate the motor voltage input to the voltage isolation amplifier. Under this condition the amplifier regulates the Silcomatic voltage to zero with the line contactor open. This type of operation is called voltage suicide.

VOLTAGE REGULATOR

By reconnecting the screw-type jumpers in the VSA voltage or Speed Amplifier, the module can be used as a voltage regulator with current limit. Figure 5 shows the simplified block diagram of this regulator.

The voltage isolation amplifier senses Silcomatic voltage. The input circuit compares the output of the isolation amplifier with the reference input and produces a voltage error which is amplified by the operational amplifier. The gain of the operational amplifier is set by the amplifier feedback circuit for the desired regulator stability. This circuit is a different circuit than the one used by the speed regulator. The current limit circuit for the voltage regulator overcomes the voltage error from the input circuit to provide motor current limit control similar to the speed regulator.

The jog circuit is identical to the speed regulator except that the jog reference sets Silcomatic voltage rather than tachometer voltage.

With the line contactor open, Silcomatic voltage is set at zero using the voltage suicide described in the "Speed Regulator" section.

OPERATING AMPLIFIER CARD (IC3600A0AC1)

The operational amplifier is a five-stage, transistorized, high-gain, inverting d-c amplifier. Both the amplifier symbol and basic connections are shown in Fig. 6. This card amplifies the voltage, V_G, which is impressed on the input from card pin 16 to common. The amplifier produces an output voltage V_O, with a polarity at pin 46 opposite that of the polarity at pin 16. For a swing in input voltage of ± 0.8 millivolt, the output voltage will swing ± 30 volts. With this high gain, the input voltage to the amplifier can be assumed to be essentially zero for purposes of description of operation.

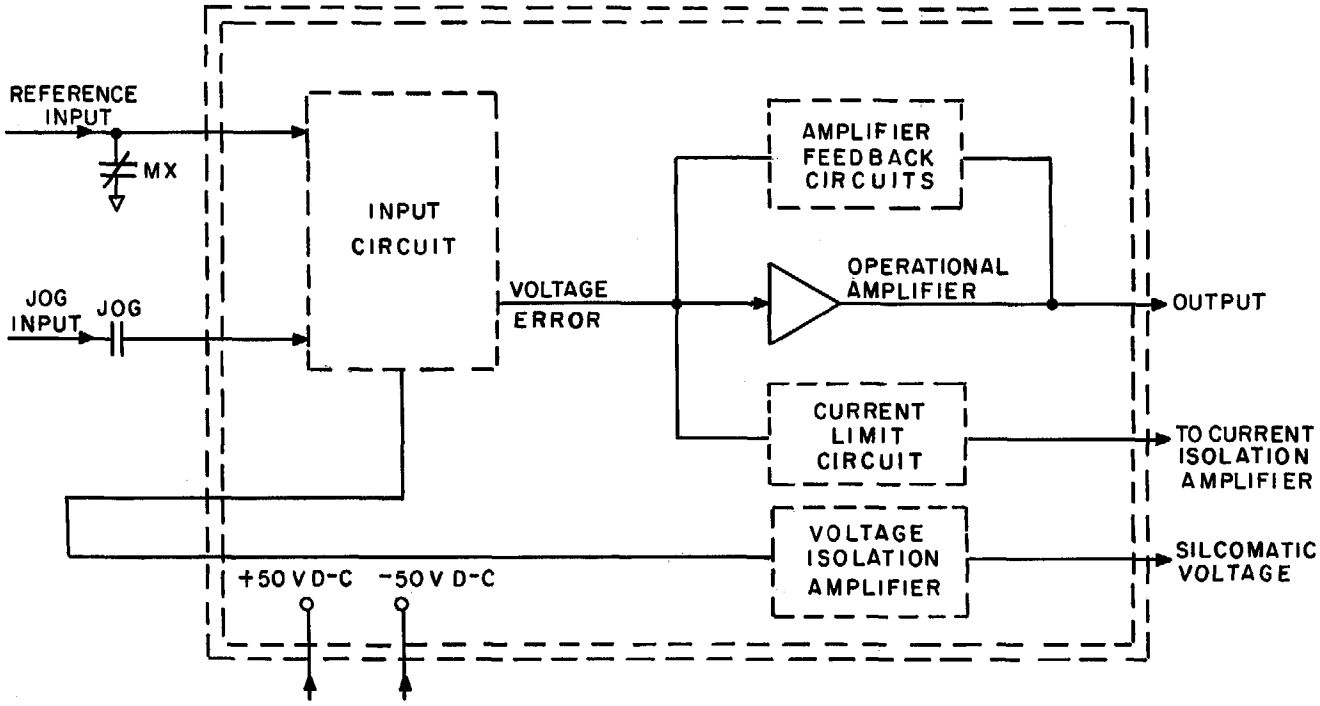


Fig. 5. Simplified diagram of voltage regulator

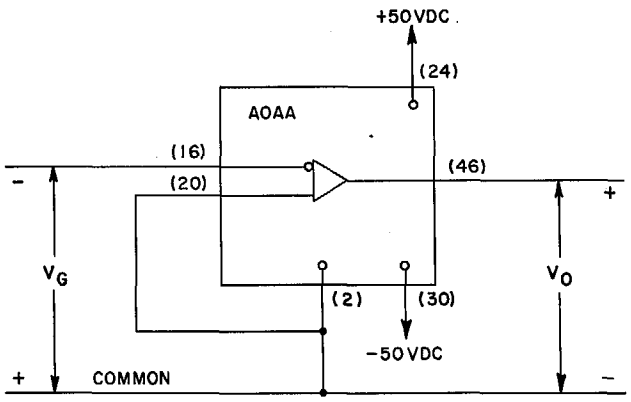


Fig. 6. Operational amplifier with printed-board connections

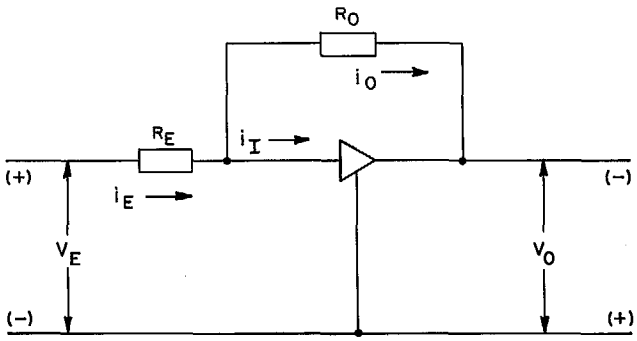


Fig. 7. Operational amplifier showing input, output, and feedback resistors

When the feedback is used around the amplifier, the gain of the amplifier is modified by the impedance of the feedback network as shown in Fig. 7. Since the amplifier input, V_G, is approximately zero and the input impedance to the amplifier is high, i_I is approximately zero; therefore, $i_O = i_E$.

Also $i_O = \frac{V_O}{R_O}$ and $i_E = \frac{V_E}{R_E}$.

Therefore, $\frac{V_O}{R_O} = \frac{V_E}{R_E}$ or $V_O = (R_O) \times \frac{(V_E)}{(R_E)}$ where:

i_E – Input current
i_O – Feedback current
V_E – Input voltage
V_O – Output voltage
R_E – Input impedance
R_O – Feedback impedance

The operational amplifier therefore amplifies the input current, i_E, by the quantity, R_O, to produce an output voltage, V_O. This output voltage, V_O, has the opposite polarity of V_E. By varying R_E, the gain of the operational amplifier can be varied. The value of R_E is set by the gain adjust potenti-

ometer in VSA. For the particular amplifier feedback circuit used in VSA, see the following sections.

Additional information on the operational amplifier including the elementary diagram, circuit characteristics and application notes is contained in the Appendix to these instructions.

ISOLATION AMPLIFIER (IC4501A101A)

The isolation amplifier is used to isolate the Silcomatic output voltage from the d-c control circuits. Contained within this amplifier is a 2.5-kilocycle oscillator connected to two amplistats for full-reversing output. As shown in Fig. 8, degenerative feedback if provided through the isolation amplifier gain potentiometer, RH24, to set the gain of the amplistat.

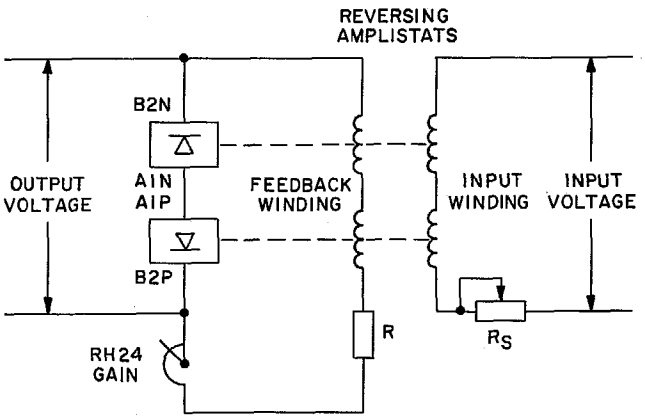


Fig. 8. Isolation amplifier, basic circuit

Resistor RS controls the input current to the isolation amplifier at rated Silcomatic voltage. For a 500-volt Silcomatic power supply, the full resistance of RS is used. The center tap on RS is provided to reduce the resistance in series with the input winding for a 250-volt Silcomatic power supply. At rated Silcomatic voltage, the output of the isolation amplifier is 15 volts, controlled by the adjustment of gain potentiometer RH24.

An internal filter attenuates the five-kilocycle output ripple of the isolation amplifier.

INPUT CIRCUIT

The input circuit is shown in Fig. 9.

For speed regulator operation, jumper N is disconnected removing the voltage isolation amplifier from the circuit, When MX is energized, the tachometer input V_T is compared with the reference input V_R through R_T and R_R respectively. For

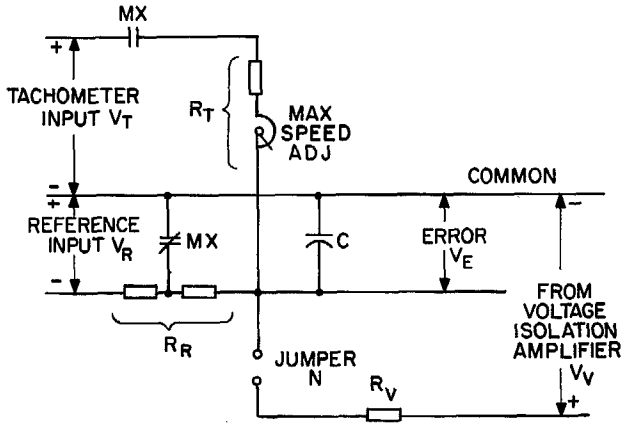


Fig. 9. Input circuit

steady-state operation, no current is required to the operational amplifier from the error voltage V_E and therefore V_E is essentially zero. With V_E at zero and no current flowing to the amplifier, the current through R_T will equal the current in R_R as follows:

$$\frac{V_T}{R_T} = \frac{V_R}{R_R} \quad \text{or} \quad \frac{V_T}{V_R} = \frac{R_T}{R_R}$$

For a given value of fixed resistors in the bridge, rotating the Maximum-Speed-Adjust potentiometer clockwise causes R_T to increase resistance; therefore, the regulated voltage V_T must increase for a fixed reference voltage V_R. The fixed portion of R_T is adjustable by means of jumpers on the module terminal board to provide for different maximum regulated voltages.

For voltage regulator operation, the tachometer input is disconnected and jumper N is made. Under this condition, V_V is compared with V_R through R_V similar to the action as described with the speed regulator input circuit and

$$\frac{V_V}{R_V} = \frac{V_R}{R_R} \quad \text{or} \quad \frac{V_V}{V_R} = \frac{R_V}{R_R}$$

Capacitor C in Fig. 9 is used as a tachometer voltage ripple filter in the speed regulator. In the voltage regulator, it is used as a compensating time constant.

Jog operation for both the speed and voltage regulator is accomplished by using the circuit shown in Fig. 10.

Closing the jog interlock applies a voltage to R_J. This voltage is adjustable by the Jog Adjust potentiometer and causes tachometer voltage or

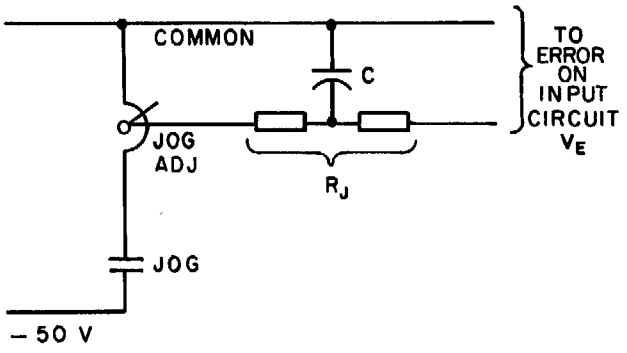


Fig. 10. Jog circuit

the output of the voltage isolation amplifier to increase.

AMPLIFIER FEEDBACK CIRCUIT

Two circuits are available for feedback around the operational amplifier for regulator stability. These circuits are shown in Fig. 11.

R1 and C1 are used for voltage regulator stability by connecting jumper W. C2, C3, R2, and the lead adjust potentiometer are used for speed regulator stability by connecting jumpers X and Y. For d-c error voltage, V_E, the full gain of the amplifier is used. In this application, the capacitor blocks d-c flow except for leakage resistance and there is no feedback. For a-c error voltages, the feedback components supply the time constants necessary to stabilize the regulator.

In addition to the stability networks around the amplifier, the two limit circuits are used to limit the positive and negative output of the amplifier. The positive limit on the amplifier operates when the input error from the reference and feedback voltages on VSA produce a positive voltage, V_O, at the amplifier output sufficient to impress 0.5 volt across CRP. When this occurs, CRP conducts to provide a low-resistance path around the amplifier. This prevents a further positive increase in V_O. The negative limit operates in the same manner with CRN conducting to limit a further increase in negative voltage at the amplifier output. The limit point at the output of the amplifier is fixed at a level slightly above that voltage required to produce 100 percent voltage on the Silcomatic output.

CURRENT LIMIT CIRCUIT

The current limit circuit sets the maximum level of motor armature current. The circuit is shown in Fig. 12.

The current limit circuit receives a voltage from the current isolator which is proportional to motor armature current. For the polarities shown, increasing voltage from the current isolator will cause the voltage on the slider of the positive current limit potentiometer to increase positive from its initial negative value set by the potentiometer slider when the voltage from the current isolator is zero. When the voltage on the slider reaches +0.5 volt, CRB will conduct. A further increase in the voltage from the current isolator will cause i_{CL} to increase positive from zero. When the voltage from the current isolator reverses polarity, the negative current limit potentiometer sets the level at which CRA conducts and i_{CL} increases negative from zero.

The current limit circuit can operate by either a reference change or increased motor load. For an increase in reference, with the polarity shown on V_E in Fig. 12, i_E increases to raise the voltage output of the operational amplifier. If motor armature current exceeds the level set by the positive current limit potentiometer, CRB conducts and i_{CL} increases to supply i_E thus reducing the net current supplied to the operational amplifier.

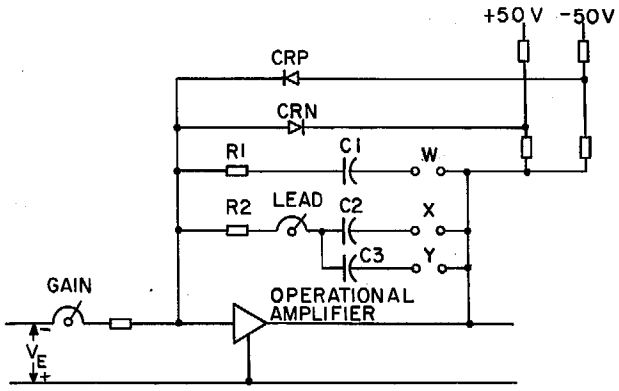


Fig. 11. Feedback circuit

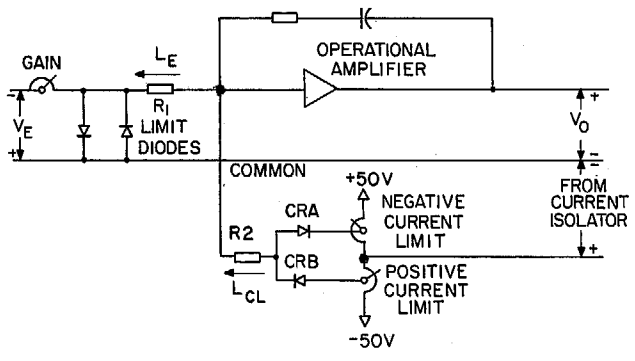


Fig. 12. Current limit circuit

The maximum value of i_E is limited by the limit diodes shown in Fig. 12. For a reference decrease, the polarity of V_E is reversed and the negative current limit potentiometer sets the level of current limit. For a load increase, the circuit operates the same way except that CRB conducts producing a current i_{CL} which causes the amplifier voltage to reduce the Silcomatic voltage.

METER AND TEST FEATURES

A 30-0-30 d-c voltmeter is provided at the output of the operational amplifier. This meter, visible from the front panel, indicates the amplifier output voltage in all modes of operation.

Test points are available at the error, operational amplifier output, current limit input, and isolation amplifier output. The test points are designated JR1, JR2, JB3, and JR4 respectively and are mounted on the voltmeter card. All of these test points can be used while the VSA voltage or speed amplifier is operating in any of its modes. All voltage measurements are made with respect to common.

In addition to the above test points available during normal operation of the VSA amplifier, a test feature is included which allows the individual cards within the amplifier to be checked on a go/no-go basis using the test points P1-P6. These test points should be used in the test position only. See "Trouble Shooting" section. When the drive is prepared for this test mode, the drive coast-stop relay is de-energized which opens the motor contactor and prevents drive operation.

INSTALLATION

No special installation instructions are required since the VSA voltage or speed amplifier is normally mounted and wired at the factory. When mounted in the field, position the module for unrestricted air flow at a height convenient for access to the controls.

INSTALLATION ADJUSTMENTS

The following adjustment procedures are those performed at installation with all drive components assembled and wired with both a-c and d-c power available. For more complete information pertaining to the operation and adjustment of specific amplifier cards, refer to the Appendix of this manual.

Selection of Voltage or Speed Regulation

The selection of voltage or speed regulation is normally made at the factory by screw-connected wire jumpers on a card located in card slot D. These jumpers are listed in Table 2 which indicates the jumpers required for a voltage regulator and the jumpers required for a speed regulator.

TABLE 2

VSA SETUP JUMPERS

Jumper	Function	Speed	Volts
A1D 32-35	Volts FB	-	1N
A1D 33-34	Volts FB	1N	-
A1D 42-45	CL Rate FB	1N	-
A1D 43-44	Amplifier FB	-	1N
A1D 47-50	Amplifier FB	1N	-
A1D 48-49	Amplifier FB	1N	-

Jumper A1D 32-35 connects the voltage isolation amplifier output to the input circuit. This jumper is removed for speed regulator operation and connected for voltage regulator operation.

Jumper A1D 33-34 connects the voltage isolation amplifier output to the operational amplifier input. This jumper is connected for speed regulator operation to provide either switch volts regulation or Silcomatic voltage suicide whenever the line contactor is open. This jumper is disconnected for voltage regulator operation.

Jumper A1D 42-45 provides a series resistor and capacitor for current limit stability in the speed regulator. It is not used for the voltage regulator.

Jumper A1D 43-44 connects the amplifier feedback circuit for the voltage regulator. It is not used for the speed regulator.

Jumpers A1D 47-50 and A1D 48-49 connect the amplifier feedback circuit for the speed regulator. They are not used for the voltage regulator.

Maximum Speed Adjustment (Speed Regulator Only)

The maximum speed of a speed regulator section is controlled by the maximum reference input voltage available and the resistance in series with the tachometer connection to the input circuit. The value of resistance in the input circuit is established by jumpers on the input terminal board A1M setting the fixed amount of resistance in addition to the maximum speed potentiometer. The selection of jumpers should be made before starting the drive. The maximum speed potentiometer is set for the desired maximum speed after the drive is started. Table 3 lists the jumpers necessary for various maximum tachometer voltage ranges with the maximum speed potentiometer giving con-

tinuous adjustment of maximum tachometer voltage within that range for a 29 volt maximum reference input.

TABLE 3

MAXIMUM TACHOMETER VOLTAGE JUMPERS

Jumper	Maximum Tachometer Volts
A1M - 2-4	82-150V
A1M - 2-3, 4-5	136-204V
A1M - 3-5	184-252V
A1M - 2-3	218-286V
A1M - 3-4	265-333V
A1M - 4-5	320-388V

For the voltage regulator, the tachometer input is not used.

Tune-up Jumpers

The selection of tune-up jumpers is made after the regulator is installed. The jumpers are listed below:

1. Current Limit Response for a Voltage Regulator. Jumper A1D 26-30 alters the amplifier feedback circuit for the voltage regulator to reduce overshoot in current limit response. This jumper is normally in the circuit when the module is shipped. It is removed to reduce current overshoot on the motor armature current going into limit. It may be left in for speed regulator operation since it has no effect.

2. Speed Regulator Stability. Jumper A1D 31-36 is normally not connected. It is connected to allow increased speed regulator gain when mechanical oscillations are present. This jumper is not connected for voltage regulator operation.

3. Current Limit Response for a Speed Regulator. Jumpers A1D 37-40 and A1D 38-39 alter the current limit circuit for the speed regulator to reduce overshoot in current limit response. These jumpers are normally not connected when the module is shipped. Add jumper A1D 38-39 to reduce current overshoot on the motor armature current going into limit. If greater reduction is required, remove A1D 38-39 and add A1D 37-40. After the current limit operation is set up, check the Silcomatic output voltage to see that the waveshape is balanced. The jumpers may be left out for voltage regulator operation since they have no effect.

Current Limit Adjustment

The current limit circuit contains two potentiometers RH1 and RH2 which set the motoring and generating current limit between 0 and 200 percent when the current isolator output is set for 7.5 volts at 100 percent full-load motor current. This adjustment is made for both the voltage and speed regulator when current limit is required. When the current limit function is not used (no current isolator), these potentiometers are set to 1.0.

Jog Adjustment

The jog speed is set by RH3 for either 0-10 percent motor speed or 0-10 percent Silcomatic voltage depending on whether VSA is set up as a speed or voltage amplifier. The jog input is used with approximately a one-second delay on application.

Voltage Isolation Amplifier Gain Adjustment

The voltage isolation amplifier gain adjustment, RH9, sets the output of the voltage isolation amplifier at +15 volts for rated Silcomatic voltage. This adjustment is required for either the voltage or speed regulator. Note that when switch volts regulation is used to precondition the amplifier, motor voltage must be zero when this adjustment is made.

Gain Adjustment

The gain adjustment, RH7, sets the response of the voltage or speed regulator. For the speed regulator, it is set for the desired speed regulator response. For the voltage regulator, it is normally set at maximum CW unless a slower response is desired.

Lead Adjustment (Speed Regulator Only)

The lead adjustment, RH6, is adjusted to cancel the effects of the major time constant of the speed regulator. The magnitude of the lead time constant is the product of the total capacitance (4 mfd made up of C6 and C7 in parallel) and the total resistance (R20 + RH6) connected in series around the operational amplifier to make up the amplifier feedback circuit. Variation of RH6 allows continuous adjustment of this lead time constant from 0.4 to 4.4 seconds. For the voltage regulator this adjustment has no effect.

TROUBLESHOOTING

GENERAL

In the event of trouble, the VSA voltage or speed amplifier has a built-in test feature. The test card (194A6785G1) mounted in slot I is removed and plugged into slot J. This de-energizes the coast-stop relay and prevents the motor line contactor from closing so that specific tests can be performed on the VSA. The circuit test points are shown in Fig. 13 and described as follows:

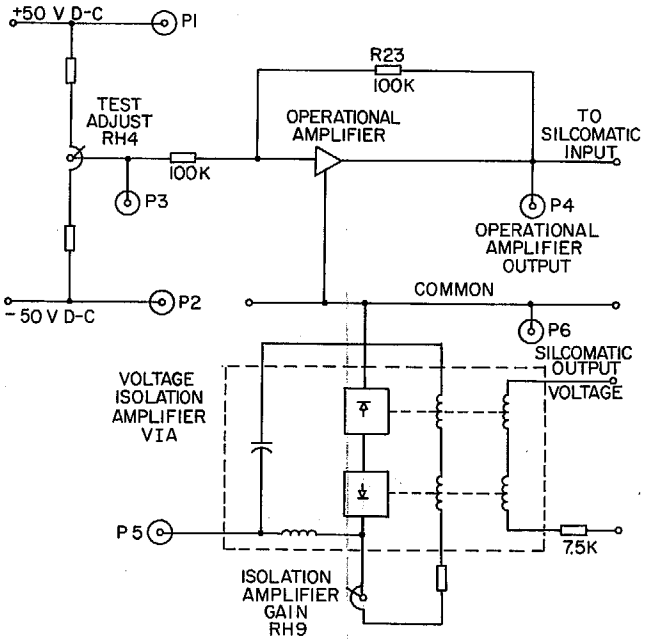


Fig. 13. Circuit test points from test card in slot J

TEST POINTS

P1 – Measures the positive 50-volt d-c supply.

P2 – Measures the negative 50-volt d-c supply.

P3 – By turning the test adjust potentiometer, RH4, the input voltage to the amplifier can be varied from +30 volts to -30 volts d-c. This test point measures the applied test voltage. Since RH4 varies the output of the VSA, this test potentiometer can be used to check the output variation of the Silcomatic unit and set RH9 for the proper gain of the voltage isolation amplifier.

P4 – The input to the operational amplifier is connected through a 100K input resistor and a 100K feedback resistor R23. These resistors provide unity voltage gain from P3 to P4. With a known test voltage at P3, the output voltage of the operational amplifier can be checked on P4 to determine that the amplifier is operating correctly. Also, the limit circuits can be checked since they control the maximum positive and negative output voltage of the amplifier.

P5 – If the output of the VSA is connected to the input of the Silcomatic by varying the test adjust potentiometer, RH4, the output of the Silcomatic can be varied. Test point P5 checks the output of the isolation amplifier since the Silcomatic unit is connected to the VSA. Note that if switch volts regulation is used, motor voltage should be zero.

P6 – Common.

PROCEDURE

All test voltages can be measured with a multimeter having a sensitivity of 20,000 ohms per volt, or greater. Typical voltages obtained by use of the test card are described in Table 4. The voltage obtained during normal operation of the amplifier are shown in Table 5.

1. Shut down drive. Allow motor to come to rest.
2. Remove test card (194A6785G1) from card slot I.
3. Insert test card in card slot J.
4. With test meter, check the following points with respect to P6 on the test card according to Table 4. Take the action indicated.

TABLE 4
TEST VOLTAGE OBTAINED FROM TEST CARD
(WHEN INSTALLED IN CARD SLOT J)

Test Point	Voltage Required with Respect to Common (P6)	Action If Voltage Not Available
P1	+50 volts, ±1 volt	1. Check external power supply. 2. Check power supply connections.
P2	-50 volts, ±1 volt	1. Check external power supply. 2. Check power supply connections.
P3	+30 to -30 volts depending on setting of test potentiometer RH4.	1. Check test card.
P4	This voltage should be equal (within ±0.1 volt) and opposite in polarity to the voltage at P3.	1. If this voltage is 30 volts or greater and (1) opposite in polarity to the test voltage, (2) reverses polarity as the test voltage polarity is reversed, replace operational amplifier. 2. If voltage does not change as test voltage is changed from +30 to -30 volts, replace operational amplifier. *NOTE: The clamp on output volts may limit positive swing of amplifier to ±26.5 volts.*
P5	The voltage should be zero for zero volts out of Silcomatic. For rated positive Silcomatic output, P6 should be +15 volts ± 0.2 volt.	1. Replace isolation amplifier.

TABLE 5
NORMAL OPERATING VOLTAGES

All voltage readings are with respect to common unless two test points are given. If two test points are given, the first one is positive.

Function	Test Point	Voltage (D-c)	Remarks
Positive Bus Voltage	A10(1)	+50 volts	
Negative Bus Voltage	A10(3)	-50 volts	
Voltage Isolation Amplifier Supply	A1Q(9)-A1Q(10)	+50 volts	
Relay Supply	A1Q(1)-A1Q(2)	+28 volts	
Input	A10(4)	0 to -29 volts	Depending on reference.
Voltage Isolation Output	JR1	0 to +15 volts	In run mode, depending on Silcomatic output voltage.
		0	In switch volts regulation.
Error	JB3	0	Steady state.
		-0.6 to +0.6 volts	In current limit.
Operational Amplifier Output	JR4	0 to +25 volts	Depending on Silcomatic voltage required.
Current Limit Input	JR2	-15 to +15 volts	Depending on value of armature current. (Note: Current input may not be used.)

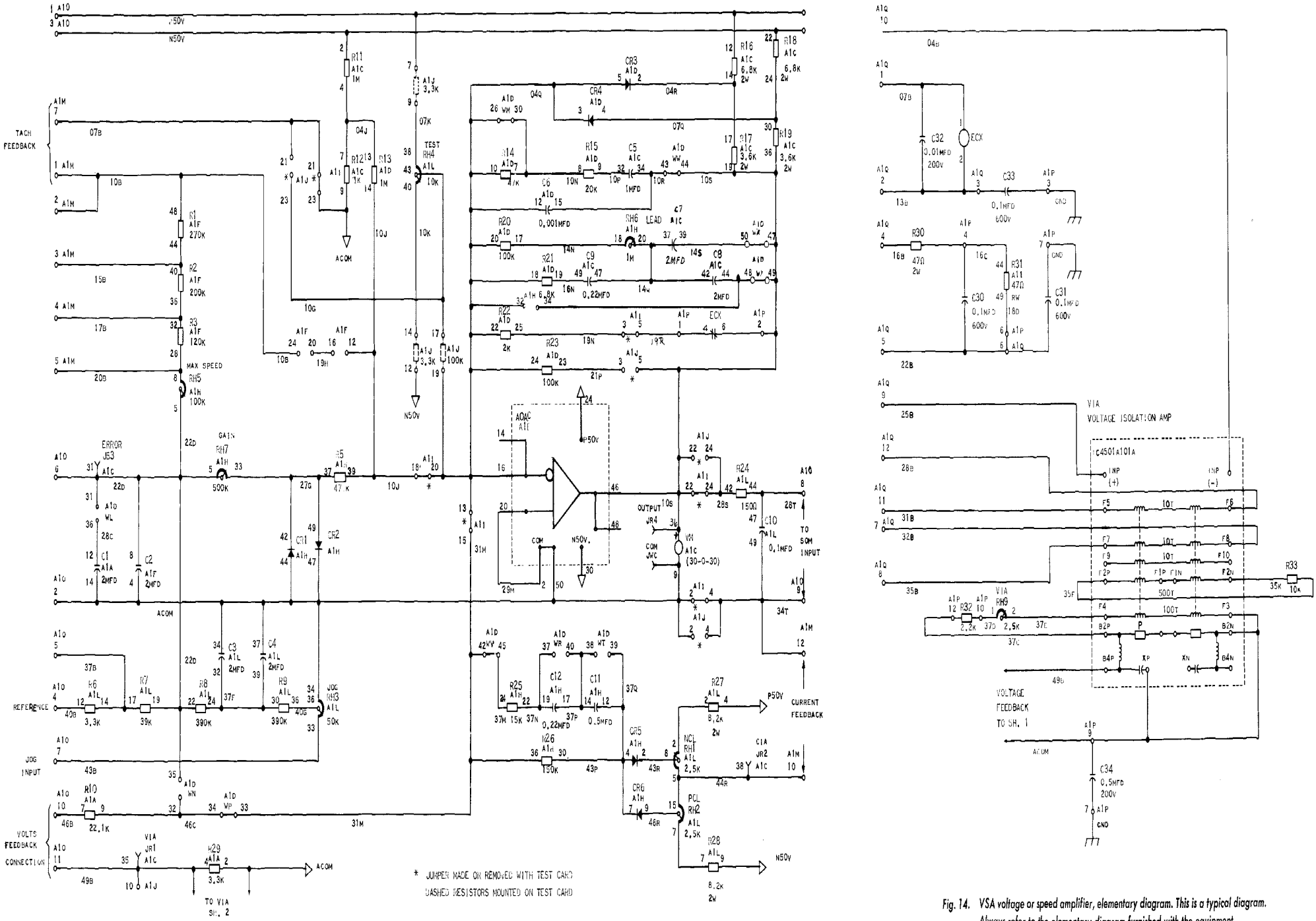


Fig. 14. VSA voltage or speed amplifier, elementary diagram. This is a typical diagram. Always refer to the elementary diagram furnished with the equipment

APPENDIX A

ISOLATION AMPLIFIER (IC4501A101A)

DESCRIPTION

The isolation amplifier consists of a 2.5-kc power supply, two full-wave bridge amplistats and two 1c filters in a common enclosure. Output terminals are provided to permit reversing or non-reversing operation with filtered or unfiltered output (see Fig. A-1).

The unit is designed primarily for isolation and use in 750-volt circuits.

CHARACTERISTICS

1 Power Requirements

Voltage	50V ±1% nominal (may be operated on 50V ±10%) absolute maximum input voltage – 55V
Current	200 ma nominal

2 Input Characteristics

	4 – 10-turn fields (connection common to both amplistats) 1 – 100-turn field (connection common to both amplistats) 1 – 500-turn field (connection individual for each amplistat)
Field Rating (max.)	16 ampere-turns each field
Common Mode Voltage	75 VDC

3 Output Characteristics

Max. Useable Output Range	
Reversing	±24V
Non-reversing	30V maximum 5V minimum
Load Impedance	
Filtered Output	5600 ohms minimum
Unfiltered Output	2700 ohms minimum
Connections	Complete reversing amplifier or two individual outputs; filtered or unfiltered. Outputs isolated from each other (100 volts) and from ground (750 volts).
Capability	Self-protected output stages to be undamaged by transients in driven inductive loads. Driving capability for similar amplistat.

4 Transfer Characteristics

Reversing – 5.6K resistive load

A. 135V/AT filtered output. 150V/AT unfiltered output

Linearity	±1 1/2% for output of ±10V ±5% for output of ±15V ±10% for output of ±20V
Max. Useable Output	±24V
Ripple	300 millivolts or less
Phase Shift	at 300 radians – 20° or less at 1000 radians – 40° or less

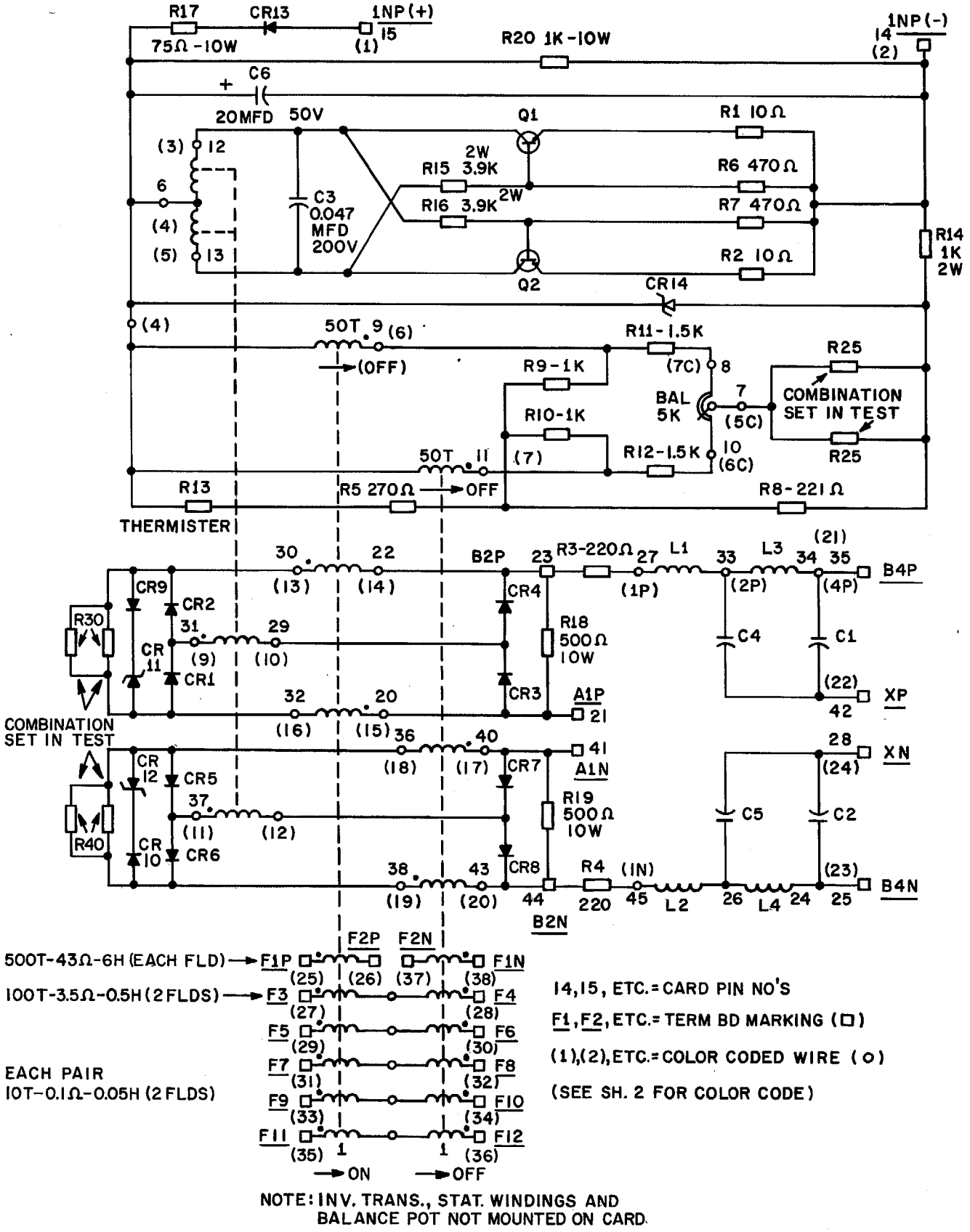


Fig. A-1. Output terminal connections for reversing or non-reversing operation with filtered or unfiltered output

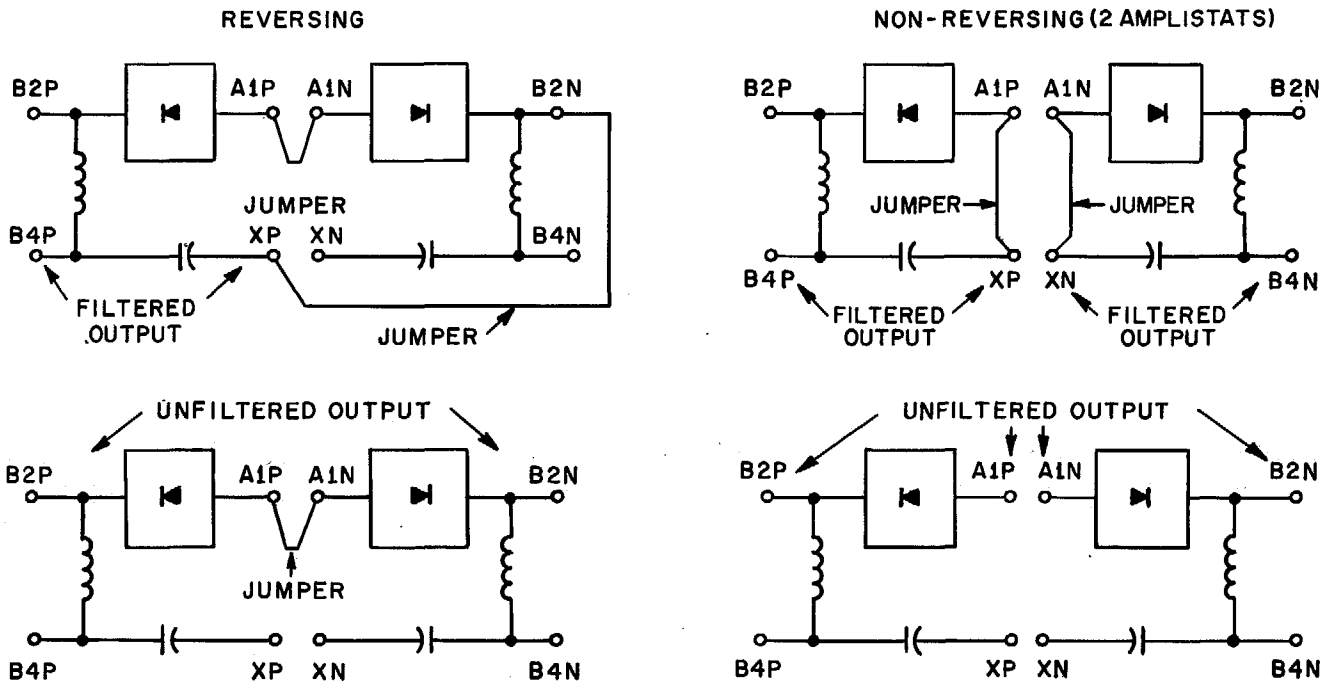


Fig. A-2. Isolation amplifier elementary diagram

A. 135V/AT filtered output. 150V/AT unfiltered output (Cont'd)

Time Constant	0.0001 + L/R
Bias Change with Time Constant	L/R = 0.025 (2 across 10T field) = -0.06AT
Offset (including offset stability)	300 millivolts (at zero and 15 volts)

B. 15V/AT with external degeneration

Max. Useable Output	±24V
Phase Shift	at 300 radians – 10° or less at 1000 radians – 30° or less 2 ohms across 10 turns at 1000 radians increases phase shift 10°

C. 1.5V/AT with external degeneration

Max. Useable Output	±15V

Individual Amplistat – 5.6K resistive load

A. 70V/AT filtered output. 75V/AT unfiltered output

Ripple	600 millivolts or less
Drift	1500 millivolts or less at rated voltage

NEMA COLOR SEQUENCE FOR CONTROL CABLES

Wire No.	Base Color	1st Stripe or Tracer	2nd Stripe or Tracer	Wire No.	Base Color	1st Stripe or Tracer	2nd Stripe or Tracer
1	Black	.....	.....	21	Orange	Green	.....
2	White	.....	.....	22	Black	White	Red
3	Red	.....	.....	23	White	Black	Red
4	Green	.....	.....	24	Red	Black	White
5	Orange	.....	.....	25	Green	Black	White
6	Blue	.....	.....	26	Orange	Black	White
7	White	Black	.....	27	Blue	Black	White
8	Red	Black	.....	28	Black	Red	Green
9	Green	Black	.....	29	White	Red	Green
10	Orange	Black	.....	30	Red	Black	Green
11	Blue	Black	.....	31	Green	Black	Orange
12	Black	White	.....	32	Orange	Black	Green
13	Red	White	.....	33	Blue	White	Orange
14	Green	White	.....	34	Black	White	Orange
15	Blue	White	.....	35	White	Red	Orange
16	Black	Red	.....	36	Orange	White	Blue
17	White	Red	.....	37	White	Red	Blue
18	Orange	Red	.....	38	Black	White	Green
19	Blue	Red	.....	39	White	Black	Green
20	Red	Green	.....	40	Red	White	Green

APPENDIX B
OPERATIONAL AMPLIFIER CARD
(IC3600AOAC1)

AMPLIFIER CHARACTERISTICS

1. Power Input	1. +50V, 24MA + load current 2. -50V, 30MA + load current		
2. Signal Input	1. $\pm$.66MV for full output range (DC) 2. $\pm$ 6.6 x 10^{-9} amps for full output range (DC)		
3. Output	$\pm$30V, $\pm$10MA (see application data)		
4. Frequency Response			
A. 10K input and feedback	285KC typical		
B. 100K input and feedback	65KC typical		
C. 1MEG input and feedback	7.6KC typical		
D. Gain Bandwidth (Open Loop)	Greater than 800KC		
5. Temperature Drift	Input Current		Input Voltage
A. 0°C to 25°C	-1.5 $\pm$2 x 10^{-9} amps/°C		$\pm$20 x 10^{-6} volts/°C
B. 25°C to 70°C	$\pm$2 x 10^{-9} amps/°C		$\pm$20 x 10^{-6} volts/°C
C. 24 hour drift (25°C)	$\pm$25 x 10^{-9} amps		$\pm$250 x 10^{-6} volts
6. Recovery from 100V Input	Recovers to 1 percent of final value in		
A. 10K input and feedback	.06 millisec. (typical)		
B. 100K input and feedback	.10 millisec. (typical)		
C. 1MEG input and feedback	.40 millisec. (typical)		
7. Recovery from Shorted Output	Recovers to 1 percent of final value in		
A. 10K input and feedback	.2 millsec. (typical)		
B. 100K input and feedback	.15 millisec. (typical)		
C. 1MEG input and feedback	.4 millisec. (typical)		
8. Output Impedance (Open Loop)	Typically less than 10 ohms from DC to beyond 1 megacycle		
9. Maximum Capacitive Load	680 PFD		
10. Frequency Limit at Maximum Amplitude			
A. Resistive load (3.3K)	70 KC (typical) (12V/microsec. Slew rate)		
B. 680 PFD load	35 KC (typical) (6V/microsec. Slew rate)		
C. 100 ohm, .01 MFD series load	3.5 KC (typical)		
11. Noise at Output (100K input and feedback)	3 millivolts due to amplifier alone.		
12. Common Mode Input	$\pm$8 volts maximum		
13. Common Mode Rejection	Can be adjusted for more than 120 DB at DC. See application notes for circuits to be used.		

APPLICATION DATA

Theory of Operation

The operational amplifier is a high-gain amplifier with a controlled "single break" frequency response. An inverting input and a non-inverting input are provided. Normally, the non-inverting input is tied to common of the analog power supply. A feedback resistor, capacitor, or series RC circuit, from output to inverting input will provide a stable closed loop which will respond in a controlled manner to any signal or signals applied to the inverting input.

Because of the high amplifier gain, the amplifier output voltage will be very nearly equal to the negative of the input voltage times the ratio of the feedback impedance to the input impedance. If the input, or feedback circuits, are networks the inverting input can be considered as being at analog common potential. The sum of the currents from input voltage to inverting input and from output voltage to inverting input can be equated to zero and the equation solved for $\frac{V_{out}}{V_{in}}$.

Effect of Temperature Variation

The amplifier is adjusted in test to have zero input voltage and current error at the temperature prevailing when the adjustment is made. This will normally be done at 25 to 35°C. When the amplifier is operated at a different ambient temperature, there will be an error voltage and current at the amplifier input because the temperature compensation circuits do not exactly compensate for the transistor characteristics. The amount of error cannot be predicted accurately unless each amplifier is pretested through the entire temperature range. The drift data given on the characteristics sheet are based on test performed on four prototype amplifiers.

The calculation shown below illustrates how the characteristic data would be used to calculate the offset band within which all amplifiers are expected to operate. Fig. B-3 shows the offset band, the calculated drift of four amplifiers based on input drifts measured in a previous test, and measured offset with this circuit.

Temperature Drift of AOAC Preamplifier

1. The amplifier input and feedback circuit with drift elements is shown in Fig. B-1.

2. The equivalent circuit is shown in Fig. B-2.

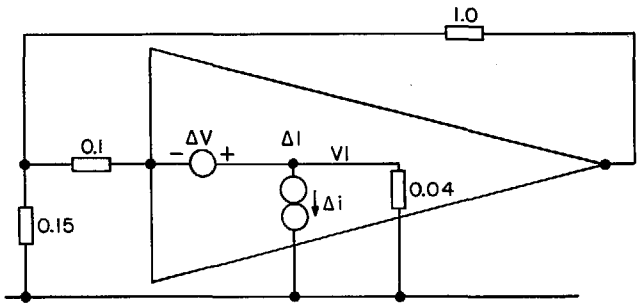


Fig. B-1. Operational amplifier input and feedback current

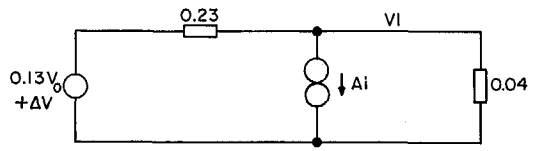


Fig. B-2. Operational amplifier equivalent circuit

3. Equations relating V_O, ΔV, and Δi for V_1 = zero.

$$\frac{.13\ V_o + \Delta V}{.23} = \Delta i$$

$$V_o = \frac{.23\Delta i - \Delta V}{.13} = 1.77\,\Delta i - 7.7\,\Delta V$$

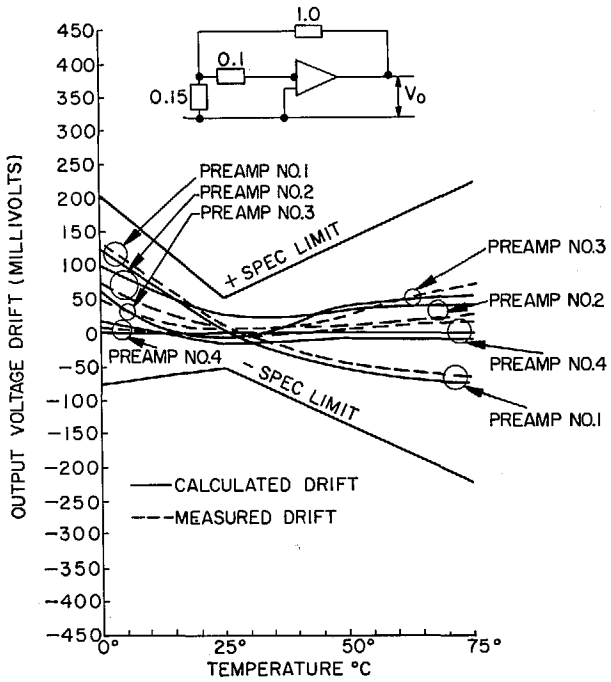


Fig. B-3. Temperature test characteristics of four prototype operational preamplifiers

If Δi is in nano-amperes and ΔV in millivolts, then V_o will be in millivolts.

If the amplifier is expected to operate over a temperature range of 25°C to 50°C, the specification sheet gives $\Delta i = \pm 25 \times 10^{-9} \pm 25 \times 2 \times 10^{-9}$ or ± 75 nano-amperes; similarly, $\Delta V = \pm 25 \times 20 \times 10^{-6}$ $\pm 250 \times 10^{-6} = \pm 750 \times 10^{-6}$ or $\pm .75$ millivolts. These values substituted in the above equation would give the maximum offset. The tracking error for two drives would be twice this value.

Noise Protection

The amplifier has an input filter, a low output impedance, bus filters and a separate Zener regulated and filtered bus system for the input stages of the amplifier. All of these features help to reduce the susceptability of the amplifier to noise transients. Large noise inputs can still cause the amplifier to saturate. To protect the amplifier from these large noise inputs it is essential that the analog bus system (especially analog common) must be stable and maintained very close to earth or enclosure potential. A direct tie from bus to enclosure should be made at one point only but distributed capacity from the common bus to enclosure will greatly reduce the effect of high frequency transients.

If a stable bus system is maintained, filters can be added which will reduce the effect of signal transients as well as high frequency ground loop currents. Some recommended filters are shown below.

1. When the amplifier output must be wired to a remote location (Fig. B-4) where it may be subjected to noise, it is recommended that a series 100 ohm resistor be mounted on the page, near the amplifier. A suitable capacitor filter should be mounted where the output leaves the controlled low noise area of the panel. The output should be carried through a twisted shielded pair to the load. It will frequently help to ground both ends of the shield if the OUTPUT LINES ARE SUBJECTED TO HIGH FREQUENCY NOISE.

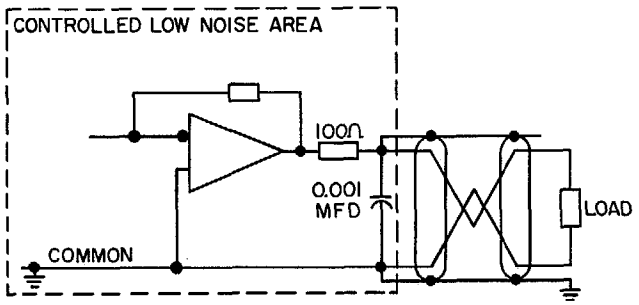


Fig. B-4. Noise filter and output shielded pair

2. Protection for input signal wires can be provided in a similar manner. It is recommended that 10 to 20 percent of the input resistance be added in series with the input line where it enters the controlled low noise area. Also a suitable filter capacitor should be mounted close to this resistor. Fig. B-5 illustrates this filtering.

This filter can be used to reduce high frequency transients. It will not be satisfactory for noise frequencies which are equal or lower than the signal frequencies to be controlled.

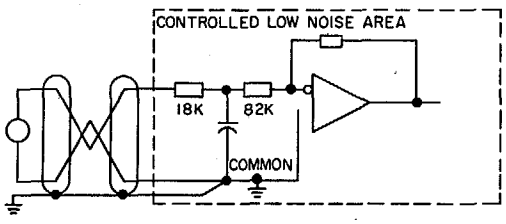


Fig. B-5. Noise filter and input shielded pair

Common Mode Rejection

DC or low-frequency common mode signals can be appreciably attenuated with a circuit as shown in Fig. B-6.

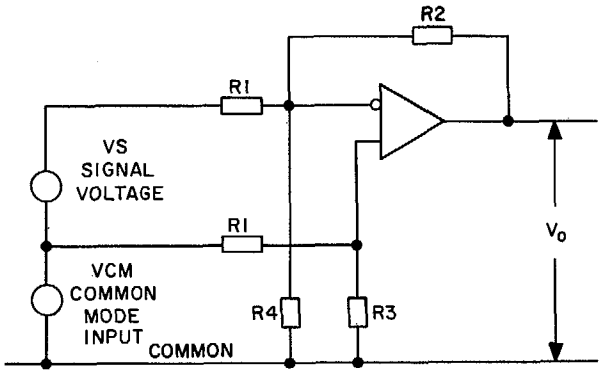


Fig. B-6. Circuit for attenuating low-frequency common-mode signals

For this configuration $V_o = -V_s \frac{R_2}{R_1}$

Also Vcm (max.) x $\frac{R_3}{R_1 + R_3} \leq 8$ volts.

R_4 is approximately $\frac{R_3 \; R_2}{R_3 + R_2}$. It is recommended that R_4 should consist of a fixed resistor and a series rheostat so that a ± 10 percent adjustment can be made. For DC common mode signals the common mode rejection can be set as high as 120 DB. At 60 cycles, the rejection ratio is reduced

because of the amplifier stabilizing is not balanced for both inputs. An additional RC circuit in the inverting input can be used to improve the high frequency common mode rejection. This requires careful selection of components and is not recommended if a high degree of AC common mode rejection is required. This circuit shown in Fig. B-7 can be easily adjusted to give up to 72 DB rejection of 60-cycle common mode voltage.

For this configuration $V_0 = -V_S \frac{R_3}{R_1}$

Also Vcm (max.) x $\frac{R_2}{R_1} \leq 30$ volts

The rejection ratio is independent of impedance in series with the common mode input. Impedance of the signal source may reduce the common mode rejection. The resistor R_1 identified by an asterisk above should be adjustable to compensate for signal source impedance.

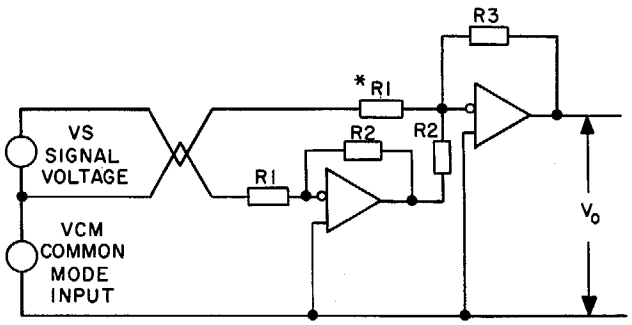


Fig. B-7. Circuit for attenuating high-frequency common-mode signals

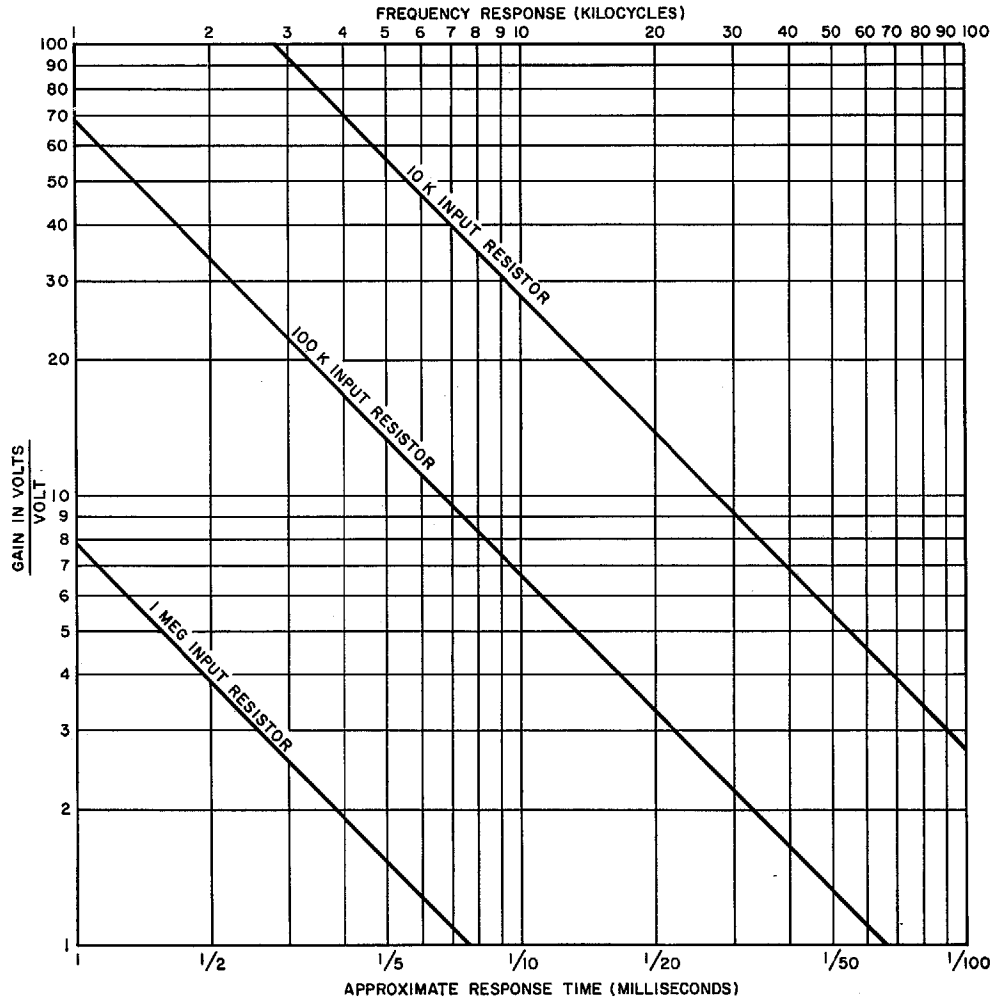


Fig. B-8. Operational amplifier typical frequency response curves. Graph also shows response time for 99 percent recovery from a step input. This information is subject to considerable error above 100 kc because of a stray capacitance of wiring and capacitance of resistors in the input and feedback circuits.

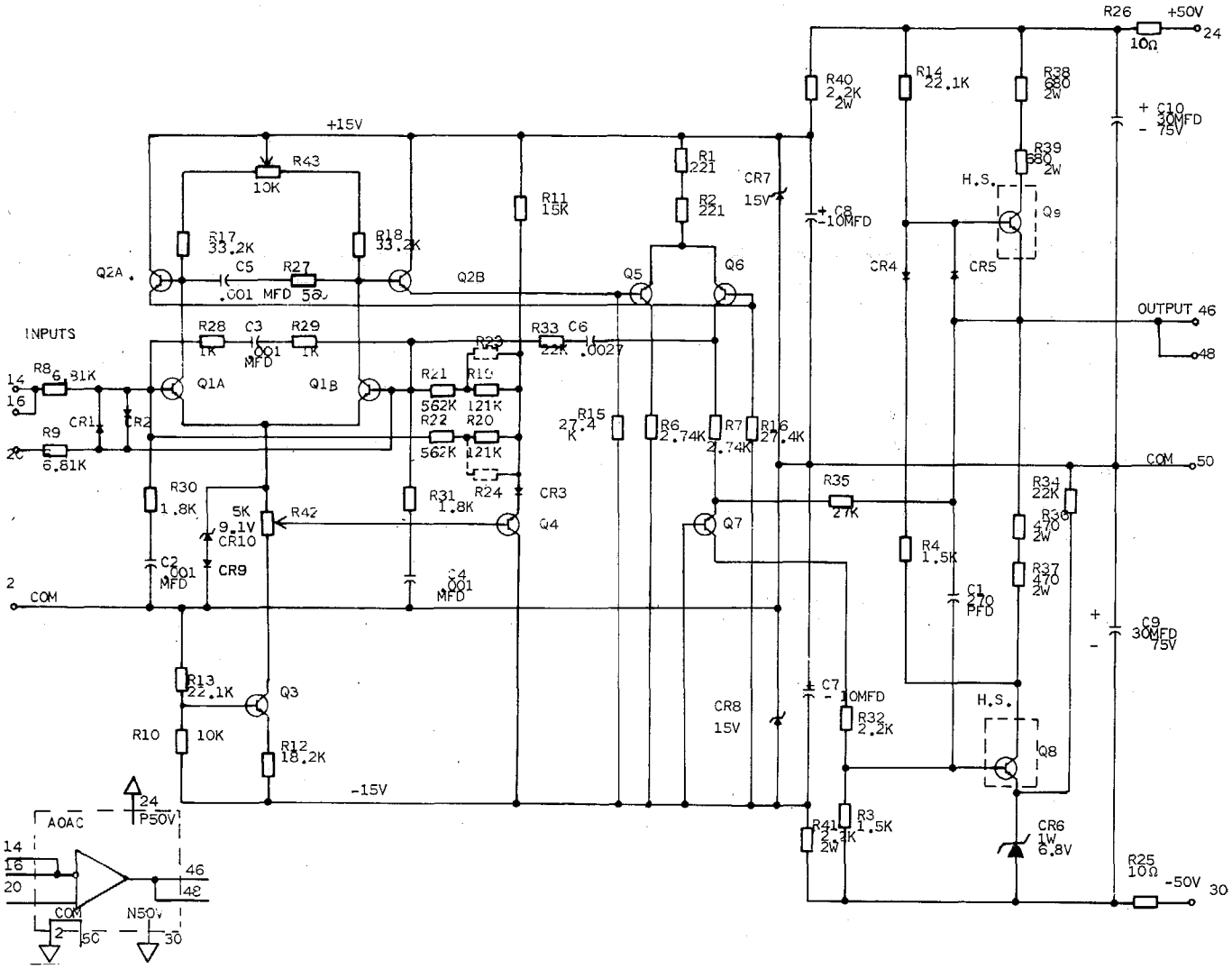


Fig. B-9. Operational amplifier (IC3600AOAC1), elementary diagram

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